



POLITECNICO
MILANO 1863

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Group 8

FINANCIAL ENGINEERING

Assignment n°4b - BS

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1 Part I: Equal Risk Contribution (ERC)

1.1 a. Implementation

The ERC portfolio allocates weights $\mathbf{w} \in \mathbb{R}^N$ so that every asset contributes equally to total volatility. By Euler's theorem:

$$\sigma(\mathbf{w}) = \sum_{i=1}^N w_i \frac{(\Sigma \mathbf{w})_i}{\sigma(\mathbf{w})} = \sum_{i=1}^N \text{RC}_i(\mathbf{w}),$$

where $\text{RC}_i(\mathbf{w}) = w_i(\Sigma \mathbf{w})_i / \sigma(\mathbf{w})$ and $\text{PCR}_i = \text{RC}_i / \sigma(\mathbf{w})$. The ERC condition $\text{PCR}_i = 1/N$ is obtained by solving:

$$\min_{\mathbf{w}} \sum_{i=1}^N \left(\text{PCR}_i(\mathbf{w}) - \frac{1}{N} \right)^2 \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \geq 0.$$

1.2 b. Sanity Checks

1.2.1 Risk-contribution dispersion and IV–ERC weight gaps

Three diagnostics are tracked monthly using the sample covariance (Figure 1):

1. **MAD** between IV and ERC weights: $\frac{1}{N} \sum_i |w_i^{\text{IV}} - w_i^{\text{ERC}}|$.
2. **Max weight gap**: $\max_i |w_i^{\text{IV}} - w_i^{\text{ERC}}|$.
3. **Normalised PCR range**: $N(\max_i \text{PCR}_i - \min_i \text{PCR}_i)$, zero at the exact ERC solution.

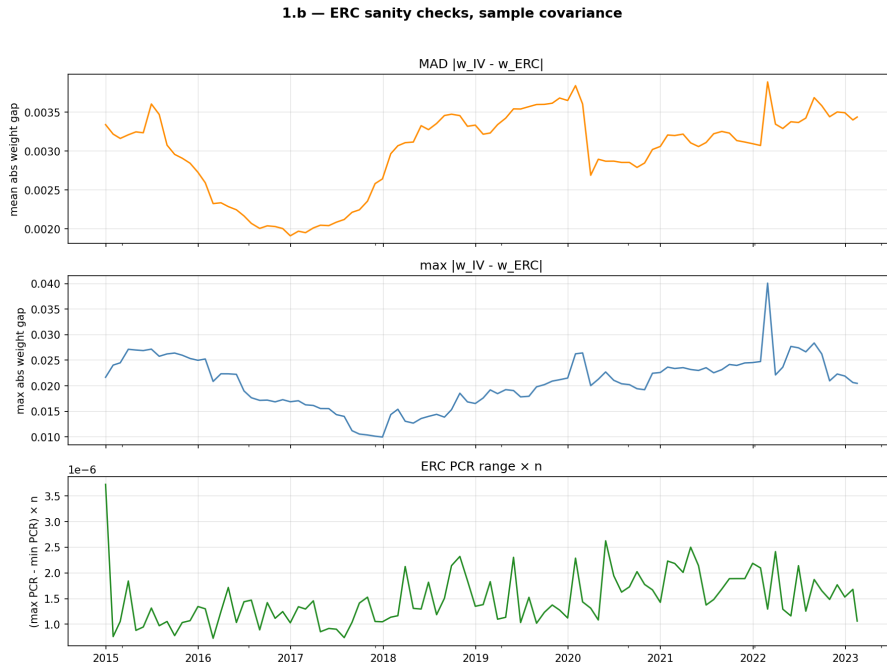


Figure 1: Sanity-check metrics over all monthly rebalances (sample covariance).

The MAD between IV and ERC weights is persistent (0.20–0.38% per asset), reflecting the structural difference between using the full Σ versus only its diagonal. The max weight gap fluctuates between 1% and 3% (brief 4% spike in 2022). The normalised PCR range stays near 10^{-6} , confirming optimizer accuracy.

Table 1: Summary of sanity-check metrics.

Metric	Mean	Max
$\text{MAD}(w^{\text{IV}}, w^{\text{ERC}})$	~ 0.0030	~ 0.0039
$\max_i w_i^{\text{IV}} - w_i^{\text{ERC}} $	~ 0.021	~ 0.040
$N(\max_i \text{PCR}_i - \min_i \text{PCR}_i)$	$\sim 1.5 \times 10^{-6}$	$\sim 3.8 \times 10^{-6}$

1.2.2 IV $\equiv$ ERC under a diagonal covariance matrix

When Σ is diagonal, $w_i^{\text{ERC}} = (1/\sigma_i)/\sum_j (1/\sigma_j) = w_i^{\text{IV}}$. Zeroing all off-diagonal entries at the last rebalance and re-running the optimizer confirms this at machine precision (Table 2).

Table 2: IV vs. ERC discrepancy on a diagonal Σ .

Metric	Value
$\text{MAD}(w^{\text{IV}}, w^{\text{ERC}})$	1.76×10^{-12}
$\max_i w_i^{\text{IV}} - w_i^{\text{ERC}} $	8.90×10^{-12}
PCR range: $\max_i \text{PCR}_i - \min_i \text{PCR}_i$	1.81×10^{-9}

1.3 c. Backtest: ERC across Three Covariance Estimators

1.3.1 Setup

Monthly end-of-month rebalancing, rolling 504-trading-day estimation window (≈ 2 years), minimum 95% data coverage per asset. Estimators: (1) Sample, (2) Const. Corr. (LW03), (3) Mkt Factor (LW02) with equally-weighted universe return as market proxy.

1.3.2 Results

Table 3: ERC backtest summary. Sharpe ignores risk-free rate; Max DD from rolling peak; Mean TO monthly one-way; Final Wealth base = 1.

Estimator	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Max DD (%)	Calmar	Final Wealth	Mean TO (%/mo)
Sample	8.89	18.68	0.48	-38.36	0.23	2.02	1.11
Const. Corr. (LW03)	8.83	18.84	0.47	-38.53	0.23	2.01	1.01
Mkt Factor (LW02)	8.88	18.67	0.48	-38.33	0.23	2.02	1.11

1.c — ERC Backtest: Sample vs Const.Corr vs Mkt Factor

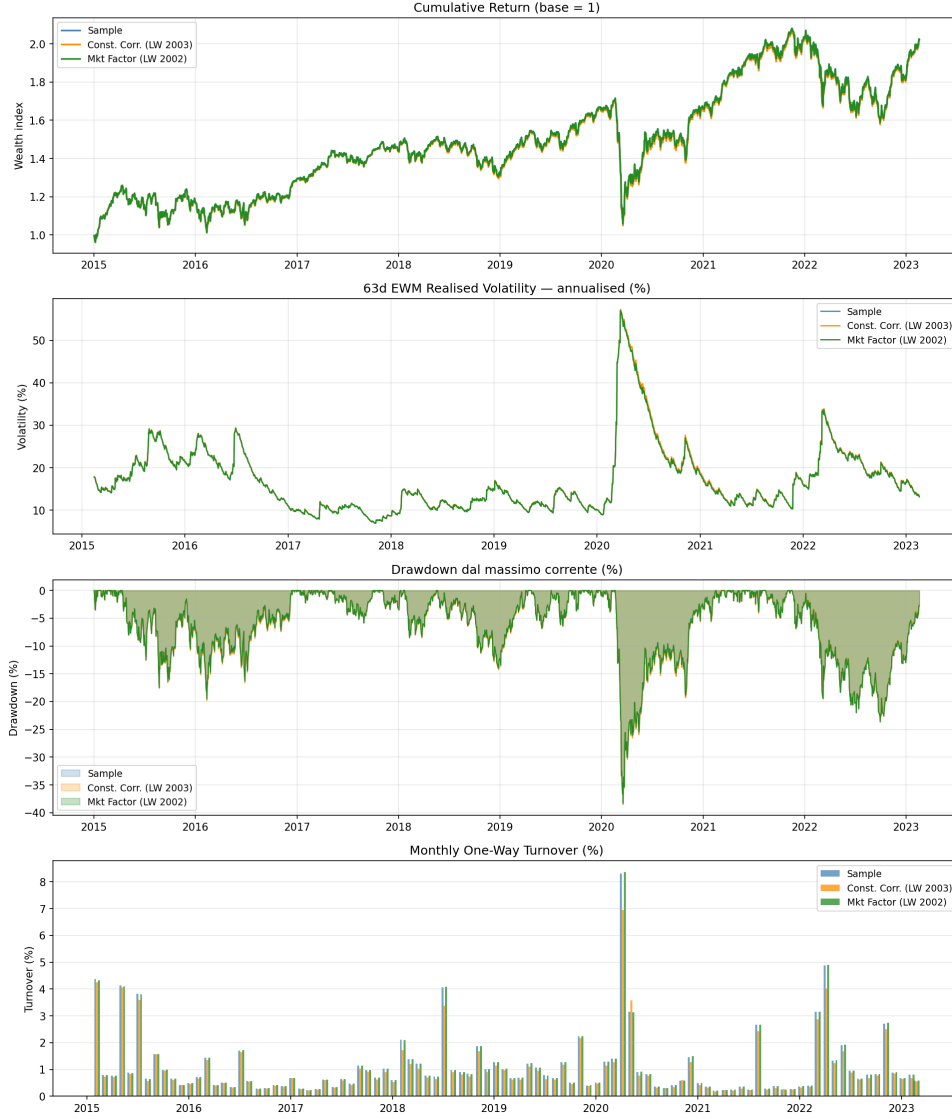


Figure 2: ERC backtest: cumulative return, 63-day EWM volatility, drawdown, and monthly turnover.

1.3.3 Discussion

Performance and Risk. The three estimators yield nearly identical results: annualized returns differ by only 6 bps, Sharpe ratios are stable at 0.47–0.48, and realized volatility paths coincide even during 2020 and 2022 shocks. Since ERC relies solely on relative covariance structure (not return forecasts), the sample matrix suffices for large-cap universes.

Drawdown. Maximum drawdowns cluster within 20 bps ($\approx -38.4\%$); the Calmar ratio is identical (0.23) across all estimators, confirming shrinkage offers no additional downside protection within ERC.

Turnover. LW03 reduces mean monthly turnover to 1.01% (vs. 1.11% for Sample and LW02, a 9% saving) by compressing correlations toward a single scalar, filtering high-frequency estimation noise. LW02 retains noisy residual variances, failing to improve over the raw sample.

Conclusion. The choice of estimator should be driven by operational efficiency rather than risk-return considerations. **LW03** is optimal, delivering consistent turnover savings at no performance cost.

2 Part II: Hierarchical Risk Parity (HRP)

2.1 a. Correlation Detoning

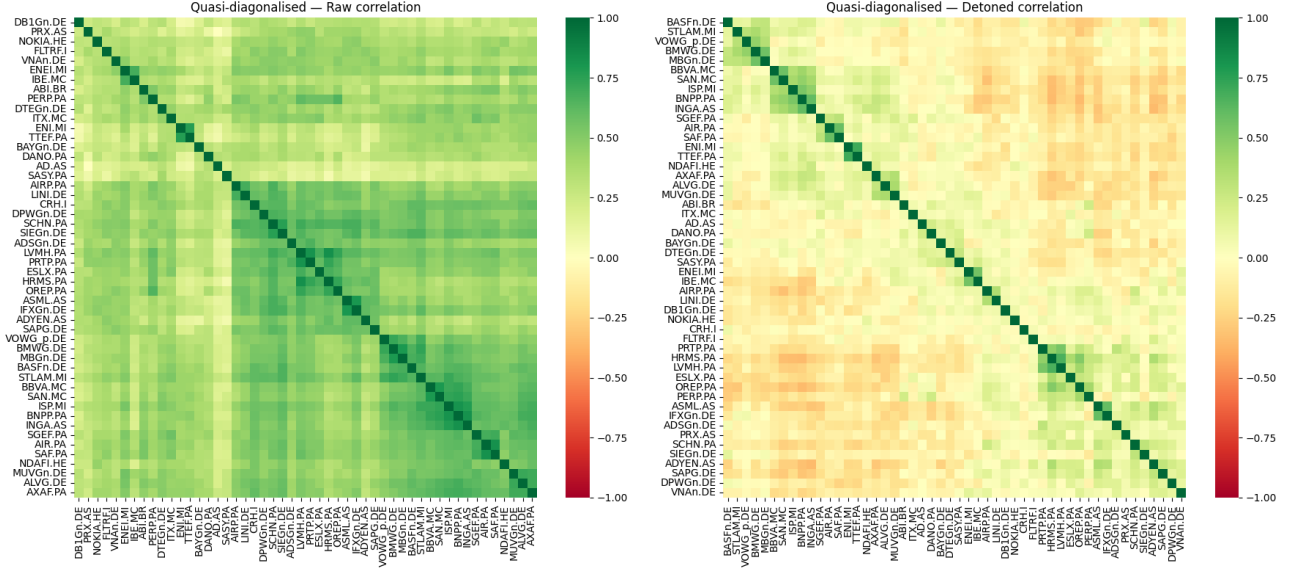


Figure 3: Correlation matrices: Raw vs. Detoned.

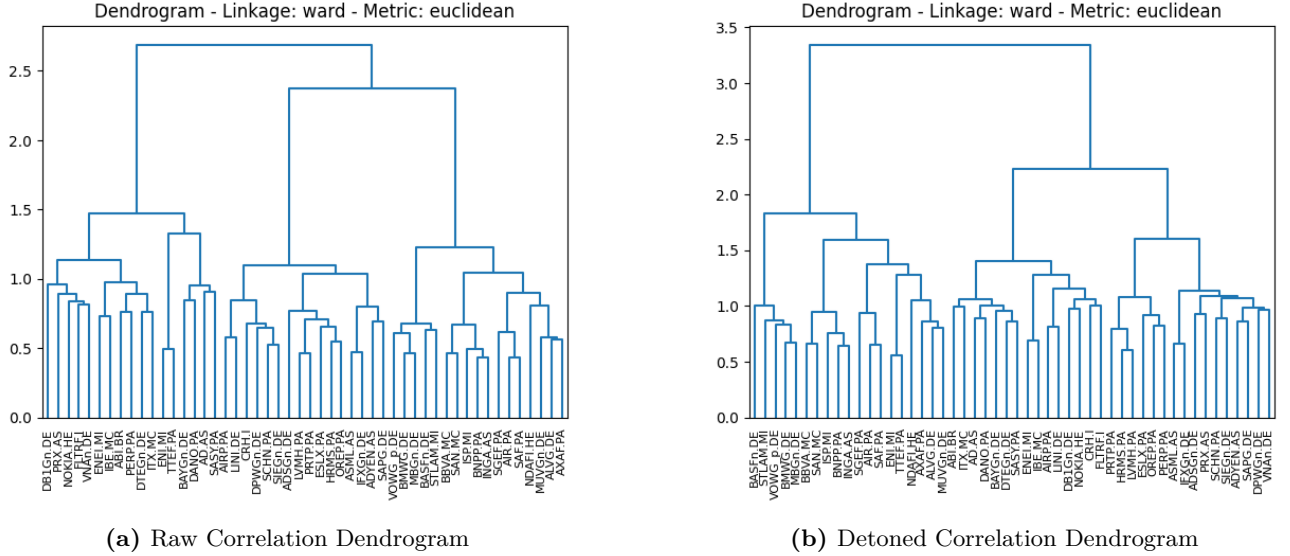


Figure 4: Dendrograms: Raw vs. Detoned.

- **Quasi-diagonalized matrices:** The raw matrix shows a broad positive correlation block across most assets. Detoning exposes a sharper block-diagonal structure, with correlated clusters standing out clearly along the diagonal and off-diagonal blocks showing near-zero (or negative) correlation, revealing diversification opportunities.
- **Dendrograms:** Removing the market factor increases the López de Prado distance (lower $\rho \Rightarrow$ larger distance), leading to deeper, more separated clusters.

2.2 b. Parametric HRP — Weight Profiles at the Last Rebalance

All 16 combinations of four linkage methods, two traversal strategies, and two detoning flags are tested (euclidean distance throughout). Concentration is measured by $\text{HHI} = \sum_i w_i^2$; EW benchmark: $\text{HHI}_{\text{EW}} = 1/50 = 0.020$.

Table 4: HRP weight statistics at the last rebalance (sample covariance, euclidean, no detoning unless noted).

Linkage	Traversal	Detoning	Min w	Max w	Std w	HHI
Single	Rec. Bisection	No	0.0057	0.0725	0.0145	0.0303
Single	Rec. Bisection	Yes	0.0041	0.0620	0.0138	0.0294
Single	Dendr. Iter.	No	0.0000	0.4331	0.0700	0.2600
Single	Dendr. Iter.	Yes	0.0000	0.2982	0.0498	0.1416
Ward	Rec. Bisection	No	0.0051	0.0633	0.0158	0.0322
Ward	Rec. Bisection	Yes	0.0050	0.0638	0.0143	0.0300
Ward	Dendr. Iter.	No	0.0012	0.1369	0.0311	0.0674
Ward	Dendr. Iter.	Yes	0.0022	0.0695	0.0134	0.0287
Complete	Rec. Bisection	No	0.0062	0.0650	0.0142	0.0299
Complete	Rec. Bisection	Yes	0.0038	0.0568	0.0130	0.0283

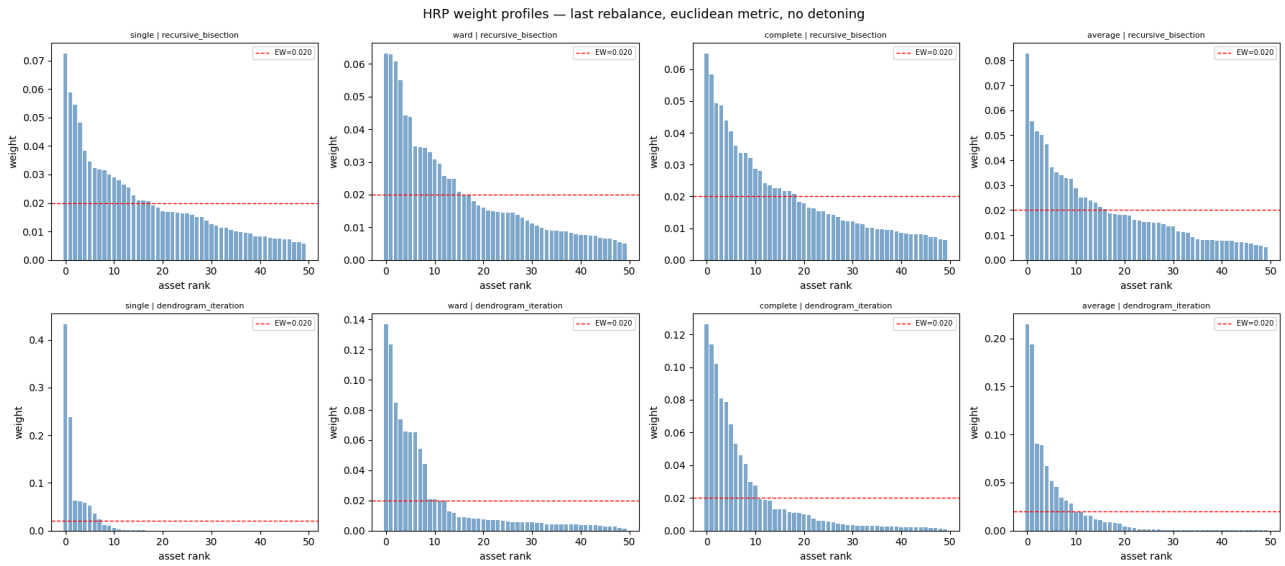


Figure 5: Sorted weight profiles for all linkage $\times$ traversal combinations (sample covariance, euclidean, no detoning). Dashed red line: $1/n = 0.020$. Top row: Recursive Bisection produces broadly similar, moderately diversified profiles. Bottom row: Dendrogram Iteration shows dramatic variation; Single linkage yields near-degenerate concentration (max $w \approx 0.43$).

Conclusions:

- **Traversal is the dominant driver.** Under Recursive Bisection, HHI clusters in $[0.029, 0.033]$ regardless of linkage. Under Dendrogram Iteration it ranges from 0.029 to 0.260 — a $13\times$ multiple of the EW benchmark. Recursive Bisection mechanically halves the sorted leaf sequence, dampening topological skew; Dendrogram Iteration follows the merge hierarchy, amplifying any tree asymmetry.
- **Single linkage + Dendrogram Iteration is degenerate.** With $\text{HHI} = 0.260$ and $\text{max } w = 0.433$, capital collapses onto two or three assets due to the chaining effect. Detoning partially mitigates this but the portfolio remains impractical.
- **Ward linkage is the most robust choice.** Ward’s balanced dendrograms limit $\text{max } w$ to 0.137 and HHI to 0.067 even under Dendrogram Iteration without detoning; adding detoning brings HHI to 0.029. Detoning has an asymmetric effect: marginal reduction (-3% to -7%) under Recursive Bisection, substantial (-43% to -57%) under Dendrogram Iteration.

2.3 c. Cluster Stability and Shrinkage Benefits

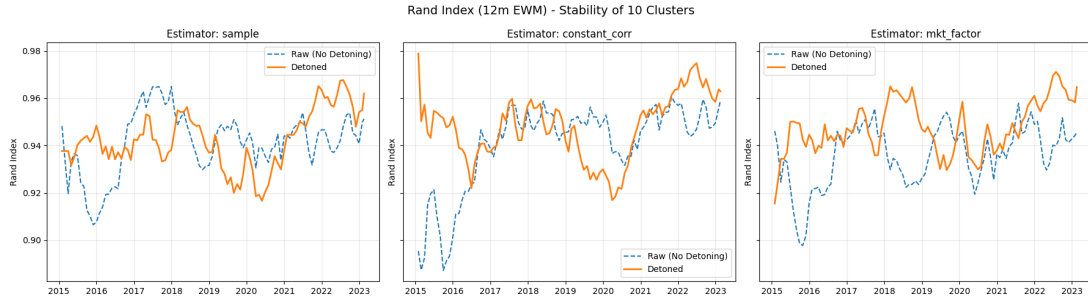


Figure 6: Cluster stability (Rand Index) for 10 clusters across estimators.

Two conclusions emerge:

- **Detoning dominates.** Across all estimators, the detoned configuration maintains a consistently higher Rand Index, confirming that removing the first principal component filters market noise that causes cluster instability.
- **Resilience under stress.** Raw stability fluctuates significantly during market stress regimes; detoned stability remains more resilient, providing a reliable foundation for the top-down allocation phase.

2.4 d. Sensitivity of HRP Weights

- **Linkage propagation: Ward linkage** achieves the best diversification by minimizing within-cluster variance, producing balanced, symmetric dendrograms. **Single linkage** yields highly concentrated portfolios due to the chaining effect.
- **Traversal comparison:** On average, **Recursive Bisection** provides better diversification (mean HHI: 0.027 vs. 0.082) and lower turnover (0.106 vs. 0.158) than **Dendrogram Iteration**. The two methods converge only when the dendrogram is symmetric; they diverge severely under Single linkage, where Dendrogram Iteration faithfully reflects the skewed structure while Recursive Bisection overrides it.

3 Part III: Connecting the Dots

3.1 a. Robustness to Estimators

- **ERC is significantly more robust.** Mean L1 distance across estimator pairs: **0.0116** for ERC vs. **0.1447** for HRP. ERC's convex optimization ensures smooth, proportional weight adjustments to small covariance changes; shrinkage mainly affects off-diagonal elements while asset variances — the primary ERC drivers — stay largely intact.
- **Topological sensitivity of HRP.** HRP avoids matrix inversion instability but introduces sensitivity to the dendrogram structure. Small changes in correlation distances can alter which cluster an asset joins; an early split in a different direction forces large capital reallocations. A minor input perturbation can thus trigger a discontinuous, large-scale weight shift.

3.2 b. Performance Comparison: ERC vs. HRP

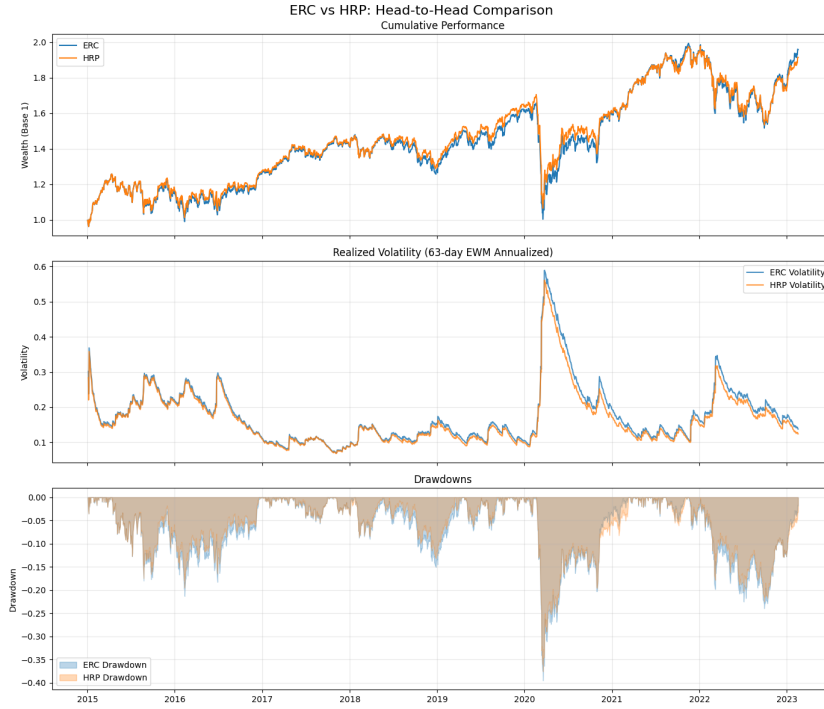


Figure 7: ERC vs. HRP head-to-head comparison.

	ERC	HRP
Ann. Return	8.89%	8.16%
Ann. Volatility	18.68%	18.13%
Max Drawdown	-38.36%	-37.56%
Mean Turnover (per rebal.)	1.11%	21.17%

Table 5: ERC vs. HRP performance summary.

- **Risk and drawdowns (HRP advantage):** HRP offers lower annualized volatility (18.13% vs. 18.68%) and a shallower max drawdown (-37.56% vs. -38.36%), confirming that clustering by correlation topology prevents over-allocation to correlated assets during market shocks.
- **Returns (ERC advantage):** ERC generates a slightly higher return (8.89% vs. 8.16%), as it does not penalize correlated assets as aggressively, retaining slightly higher weights in momentum-driven sectors during bull markets.
- **Turnover (decisive ERC advantage):** ERC requires only 1.11% monthly turnover vs. HRP's 21.17%. Normal fluctuations in the rolling covariance matrix can alter cluster topologies, forcing massive capital reallocations at each HRP rebalance. In a real-world setting with transaction costs, HRP's turnover would severely erode its gross performance advantage.

3.3 c. Shrinkage Sensitivity Analysis

Across this assignment, switching estimators has had negligible impact on portfolio outcomes — in contrast to Lab 3, where shrinkage reduced turnover by up to $2.5\times$ (at the COVID peak) and condition numbers peaked near 400 for the sample matrix.

Structural explanation. The difference traces to matrix inversion. Optimization-based strategies compute $\mathbf{w}^* \propto \hat{\Sigma}^{-1}\mathbf{v}$, amplifying noise proportionally to $\kappa(\hat{\Sigma})$; shrinkage breaks this chain. Risk-based strategies sidestep inversion:

- **ERC** minimizes $\sigma(\mathbf{w}) = \sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}}$; since shrinkage leaves asset variances (the primary driver) largely intact, weights barely shift.
- **HRP** uses $\hat{\Sigma}$ only to extract a correlation matrix for clustering and then allocates by $1/\hat{\sigma}_i^2$ within clusters — both steps robust to off-diagonal perturbations by construction.

Role of the N/T ratio. With $N = 50$ and $T \approx 504$, the ratio $N/T \approx 0.10$ is low enough for the sample covariance to be well-conditioned, keeping shrinkage intensity α small and further limiting the scope for estimator differences to propagate into weights.

Conclusion. Shrinkage is most valuable when a strategy inverts $\hat{\Sigma}$. ERC and HRP operate on $\hat{\Sigma}$ directly or on derived quantities that are inherently stable, making the estimator choice secondary to the allocation mechanism, detoning, and rebalancing frequency.

4 Errors and notes in the code:

4.1 Ex4b-notebook:

Definition of parameters: estimation_window was 1 years instead of 2 as requested by the Assignment.

4.2 backtest

Align ptf and returns: substituted bfill with ffill.

4.3 hierarchical risk parity

correlation to hrp distance: in the distance formula there was a 2 instead of a $\frac{1}{2}$.

top down allocation: the initial formula for alpha1 was totally wrong, and so the relation between alpha1 and alpha2.